ಬಿ.ಎಂ.ಎಸ್. ತಾಂತ್ರಿಕ ಮತ್ತು ವ್ಯವಸ್ಥಾಪನಾ ಮಹಾವಿದ್ಯಾಲಯ

BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT

(Autonomous Institution Under VTU)

Yelahanka, Bengaluru -560119

A Major-project Phase 1 Work Report

on

"Travel Yantra: An AI Based Personalized Travel Itinerary Planner and a Virtual Tour Guide with Multilingual Support"

Submitted in partial fulfillment for the award of

BACHELOR OF ENGINEERING

in

INFORMATION SCIENCE AND ENGINEERING

by

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CERTIFICATE

This is to certify that the Major-Project work (BCS606) entitled "**Travel Yantra: An AI Based Personalized Travel Itinerary Planner and a Virtual Tour Guide with Multilingual Support**" is a bonafide work carried out by **Mr. Nithin G (1BY23IS140), Mr. Pushkar Reddy S (1BY23IS167), Mr. Rishab Paul (1BY23IS175), Mr. Rohan Balu (1BY23IS177)** in partial fulfillment for the award of **Bachelor of Engineering Degree in Information Science and Engineering** of the Visvesvaraya Technological University, Belagavi during the year 2025–26. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in this report. The Major-project phase 1 report has been approved as it satisfies the academic requirements with respect to Major-project work for the B.E Degree.

Signature of the Guide (Prof. Shwetha T)	**Signature of the Coordinator (Dr. Kalai Vani Y S)**	**Signature of the Cluster Head (Dr. N. Rakesh)**	**Signature of the HOD, ISE (Dr. Surekha K B)**

ABSTRACT

Context: Modern travel planning in India is fragmented across booking sites, review portals and map applications, forcing travellers to consult four to five disconnected platforms and spend six to ten hours per trip on logistics alone. At the same time, the country has twenty-two official languages and over seven hundred million people who are more comfortable in their regional language than in English, yet almost every mainstream travel artificial intelligence tool is English-only. This combination of platform fragmentation and linguistic exclusion is the broader context in which this work is positioned.

Problem Identified: Existing travel platforms operate in silos and offer no unified system that brings together AI-driven itinerary generation, real-time booking, on-trip cultural guidance and multilingual interaction in a single, accessible product. Generic large language models such as GPT-4 perform poorly on complex multi-constraint travel plans (less than one percent pass rate on the TravelPlanner benchmark), and Indic voice and translation pipelines underperform on travel vocabulary and code-switched queries common to Indian users.

Objective: To design and implement Travel Yantra, an AI-powered, multilingual, multi-channel travel itinerary planner and virtual tour guide for the Indian tourism context, combining specialised large language model agents, multilingual natural language processing, real-time booking integration and a virtual cultural guide, all delivered through web, WhatsApp and voice channels.

Methods: The system is built around four specialised agents (Itinerary, Accommodation, Transportation and Cultural Guidance) orchestrated by n8n. Each agent is paired with the large language model best suited to its sub-task: GPT-4 for constraint reasoning, Claude 3.5 Sonnet for cultural narrative and Gemini 1.5 Pro for real-time multimodal data lookup. The natural language stack uses IndicTrans2 for translation across five Indian languages, MURIL for code-switched understanding and Whisper for speech recognition. Cultural responses are grounded through Retrieval-Augmented Generation over a multilingual knowledge base. Routing uses Dijkstra and A*, recommendations use Neural Collaborative Filtering, and the backend uses Supabase (PostgreSQL with Row-Level Security and JWT authentication). User access is provided through a React.js web application, the WhatsApp Business API and Vapi voice AI.

Results: The proposed framework targets ninety percent or higher user satisfaction on generated itineraries, eighty-five percent or higher translation accuracy across the supported languages, sub-three-second P95 response latency, and a sixty percent reduction in user planning time, all delivered within an end-to-end project budget of twenty-five thousand rupees on a fully open-source-first stack. By combining a multi-agent architecture with multilingual NLP and

multi-channel delivery, the system is expected to bring intelligent travel planning to over seven hundred million regional language users currently excluded by English-only tools.

Keywords: Multi-Agent Large Language Model Systems, Multilingual Natural Language Processing, IndicTrans2, MURIL, Retrieval-Augmented Generation, Voice AI, WhatsApp Business API, n8n, Travel Itinerary Planning, Virtual Tour Guide, Tourism in India, Supabase, Chain-of-Thought Prompting.

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By,

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Declaration

We, hereby declare that the major-project (BCS606) titled "**Travel Yantra: An AI Based Personalized Travel Itinerary Planner and a Virtual Tour Guide with Multilingual Support**" is a record of original major-project work undertaken for partial fulfillment of Bachelor of Engineering in Information Science and Engineering of the Visvesvaraya Technological University, Belagavi during the year 2025–26. We have completed this major-project work under the guidance of **Prof. Shwetha T, Assistant Professor**.

We also declare that this major-project report has not been submitted for the award of any degree, diploma, associateship, fellowship or other titles anywhere else.

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1 INTRODUCTION

Travel and tourism contribute close to seven percent of India's gross domestic product, and the country attracts millions of domestic and international travellers every year. Despite this scale, the everyday experience of planning a trip in India remains surprisingly painful. A traveller who wants to plan even a short five-day trip is typically forced to open one website to search for flights, another for hotels, a third for attractions, a map application to check whether the chosen hotel is anywhere near the chosen sights, and then a messaging application to ask a friend or relative who has visited the destination before. Even after all of this, the final plan often feels uncertain. Studies in the popular travel domain estimate that travellers spend between six and ten hours just to assemble a single trip across these disconnected platforms. That is effectively a full working day lost on logistics, before the trip has even started.

The second, and equally important, problem is language. India has twenty-two official languages, and an estimated seven hundred million people in the country are more comfortable speaking in their regional language than in English. Most modern travel artificial intelligence tools, including chat-based assistants such as ChatGPT and Gemini, are designed primarily for English. Even where Indic language support is technically present, the quality on travel-specific vocabulary, named places, festival names and code-switched queries (Hinglish, Kanglish and similar) is noticeably weaker. As a result, a large section of the Indian travel market is effectively excluded from the benefits of recent advances in conversational artificial intelligence.

The arrival of large language models, retrieval-augmented generation, multi-agent orchestration platforms and modern multilingual neural machine translation has, however, made it possible to address these two problems together. Travel Yantra is conceived as that integrated answer. It is an AI-powered, multilingual, multi-channel travel itinerary planner and virtual tour guide built specifically for the Indian context. The system combines four specialised large language model agents, a multilingual natural language stack centred on IndicTrans2 and MURIL, retrieval-augmented cultural guidance, voice access through Vapi, messaging access through the WhatsApp Business API and a web application, all coordinated through n8n and supported by a Supabase backend.

The remainder of this report is organised as follows. Section 2 sharpens the problem statement and shows why the current state of practice is inadequate. Section 3 sets out the objectives in measurable terms. Section 4 surveys twenty-two recent research papers spanning multi-agent travel planning, multilingual natural language processing, retrieval-augmented generation, voice recognition, recommendation systems, route optimisation and tourism technology. Section 5 distils the resulting research gap and the technical challenges Travel Yantra has to solve. Section 6 presents the proposed methodology, including system architecture, agent design, data flow and tooling. Section 7 records the panel feedback received in Phase 1 Review 1 and the self-initiated refinements carried into Phase 1 Review 2. Section 8 lists the expected outcomes and Section

9 concludes the report. Section 10 lists all references in IEEE format.

2 PROBLEM STATEMENT

Modern travel planning in India is fragmented across multiple disconnected platforms, and there is no unified system that combines AI-driven itinerary generation, real-time booking, on-trip cultural guidance and multilingual interaction in a single product. Generic large language models cannot reliably plan complex multi-constraint trips, and existing Indic language pipelines remain weak on travel vocabulary and code-switched queries. As a consequence, travellers either burn several hours stitching together their own plans across four or five tools or they fall back on rigid template-based bots and cookie-cutter recommendations that ignore individual preferences, budget, group composition, language and culture.

The problem this project addresses, in concrete terms, is the absence of a single platform that can: take a user's travel intent in their own language and channel of choice, decompose it into a feasible day-by-day plan, attach real bookable links for transport and accommodation, route the user efficiently between points of interest, and act as a knowledgeable cultural companion at the destination, all without crossing the privacy and compliance constraints of Indian users.

3 OBJECTIVES

The objectives of the Travel Yantra project are deliberately measurable and time-bound, so that they can be evaluated quantitatively at the end of Phase 2.

- **AI-Powered Personalised Itinerary Generation:** Design and implement a multi-agent AI system that generates day-by-day itineraries from user preferences, budget, dates and group composition, targeting a user satisfaction rate of ninety percent or higher.
- **Multilingual Natural Language Support:** Build NLP pipelines that cover Hindi, Tamil, Telugu, Kannada and Bengali alongside English using IndicTrans2 and MURIL, with a translation accuracy target of eighty-five percent or higher and cultural context preservation across these languages.
- **Multi-Channel Deployment and Accessibility:** Deploy the system across web, WhatsApp and voice with platform availability of ninety-five percent or higher and a P95 response time of three seconds or less for ninety-five percent of queries.
- **Real-Time Booking and Route Optimisation:** Provide live booking links for flights, hotels and attractions and apply Dijkstra and A* for multi-stop route optimisation, with the goal of reducing the user's planning time by sixty percent compared to existing fragmented workflows.

- **Virtual Cultural Tour Guide:** Build a Retrieval-Augmented Generation based interactive guide for history, food, cultural significance and emergency information, with sub-three-second response latency.
- **Privacy and Cost Constraints:** Achieve all of the above on an open-source-first stack within a project budget of twenty-five thousand rupees, while remaining compliant with the Indian Digital Personal Data Protection Act 2023.

4 LITERATURE SURVEY

4.1 Overview

The literature relevant to Travel Yantra spans multi-agent large language model planning, multilingual natural language processing for Indian languages, retrieval-augmented generation, automatic speech recognition, recommendation systems, classical and heuristic route optimisation, and digital transformation in tourism. Twenty-two papers were reviewed in IEEE format and are summarised below in seven thematic groups. The strengths and limitations of each are highlighted, with explicit pointers to which component of Travel Yantra they justify.

4.2 Large Language Model Planning and Multi-Agent Systems

Xie et al. [1] introduced TravelPlanner, a benchmark of one thousand two hundred and twenty-five real travel planning queries, and reported that even GPT-4 achieves less than one percent final pass rate on complex multi-constraint plans. This is the strongest single piece of evidence in the literature that a generic single-LLM approach is not sufficient for real travel planning. Singh et al. [2] responded with Travel Optix, a CrewAI-style multi-agent system that produces itineraries in under five minutes and outperforms standalone GPT-4 and Gemini, validating the multi-agent direction but leaving multilingual support and voice access as open problems.

Brown et al. [3] introduced GPT-3 and the few-shot prompting paradigm, while Devlin et al. [4] introduced BERT and the bidirectional pre-training paradigm. Both are foundational to every modern conversational system, including Travel Yantra. Wu et al. [6] proposed AutoGen, a framework for orchestrating multiple agents in conversation, and Yao et al. [7] proposed ReAct, which interleaves reasoning and acting and is the prompting style used by the planning agents in this work. Wei et al. [15] demonstrated that Chain-of-Thought prompting elicits step-by-step reasoning in large models and is directly used by the Itinerary Agent for constraint reasoning.

4.3 Multilingual and Indic Natural Language Processing

Gala et al. [5] introduced IndicTrans2, a state-of-the-art neural machine translation system covering all twenty-two scheduled Indian languages with BLEU scores in the twenty to thirty plus range on benchmark sets. This is the single biggest technical lever for the multilingual goals of Travel Yantra. Khanuja et al. [16] introduced MURIL, a BERT-style encoder pre-trained on seventeen Indian languages, which is used in the proposed system specifically to handle code-switched queries such as Hinglish and Kanglish. Vaswani et al. [10] provided the Transformer architecture that underlies all of the above models.

4.4 Retrieval-Augmented Generation and Voice

Lewis et al. [13] proposed Retrieval-Augmented Generation, in which a generator is conditioned on documents retrieved from an external store, and showed that this materially reduces hallucination compared to parametric-only models. The Cultural Guidance Agent in Travel Yantra adopts this approach over a multilingual cultural knowledge base. Radford et al. [14] introduced Whisper, an automatic speech recognition model trained on six hundred and eighty thousand hours of weakly supervised audio that performs well on multilingual and accented speech, and is used as the speech recognition backbone behind the Vapi voice channel.

4.5 Recommendation, Routing and Evaluation

He et al. [8] introduced Neural Collaborative Filtering, a deep learning successor to matrix factorisation that is used in this work for accommodation and attraction recommendation. Dijkstra [12] provided the classical shortest-path algorithm used as the baseline routing engine, and Hart, Nilsson and Raphael [18] introduced A*, the heuristic-guided alternative used for multi-stop scheduling under time and budget constraints. Papineni et al. [11] proposed BLEU and Banerjee and Lavie [17] proposed METEOR, which together provide the translation evaluation metrics for the multilingual targets in Section 3.

4.6 Tourism Technology and Conversational Agents

Buhalis and Sinarta [9] provided the framework of real-time co-creation in tourism that motivates the choice of an interactive on-trip cultural guide. Tussyadiah [19] surveyed AI and robotics in tourism and concluded that personalisation and multilinguality remain unsolved problems. Calvaresi et al. [20] performed a systematic review of conversational agents in tourism and reached a similar conclusion, providing strong external validation of the gap that this project targets.

4.7 Frameworks and Open-Weight Models

Chase et al. [21] released LangChain, an open-source framework for composing LLM applications, used in this work for prompt and chain management around the agents. Touvron et al. [22] released LLaMA 2, an open-weight chat-tuned family of models that is held in reserve as a fall-back option in case any closed API agent in Travel Yantra becomes cost-prohibitive at production scale.

Paper	Year	Technique	Strengths	Limitations
Xie [1]	2024	GPT-4 / Gemini on TravelPlanner benchmark	Realistic 1,225-query benchmark; clear failure analysis	GPT-4 achieves $<1\%$ pass on complex plans; benchmark only
Singh [2]	2024	CrewAI multi-agent travel planner	Sub-5 min planning; beats single LLMs	No multilingual or voice support
Gala [5]	2023	IndicTrans2 NMT	BLEU 20–30+ across all 22 Indian languages	Travel-specific vocabulary needs fine-tuning
Khanuja [16]	2021	MURIL encoder for 17 Indic languages	Strong code-switched understanding	No travel domain pre-training
Wu [6]	2023	AutoGen multi-agent framework	Generic multi-agent orchestration	Not optimised for travel or Indian context
Yao [7]	2023	ReAct (reason + act) prompting	Tool use with explicit reasoning	Latency overhead per step
Wei [15]	2022	Chain-of-Thought prompting	Improves multi-step reasoning	Long prompts; cost overhead
Lewis [13]	2020	Retrieval-Augmented Generation	Reduces hallucination; grounded answers	Retrieval quality is the bottleneck
Radford [14]	2023	Whisper ASR	Robust multilingual speech recognition	Weaker on heavy Indic accents
He [8]	2017	Neural Collaborative Filtering	Strong on sparse recommendation	Cold-start problem for new users
Dijkstra [12]	1959	Shortest-path graph algorithm	Optimal, well understood	No heuristics; slow on large graphs
Hart [18]	1968	A* heuristic search	Faster than Dijkstra with good heuristic	Heuristic must be admissible
Buhalis [9]	2019	Real-time co-creation in tourism	Strong framework for on-trip services	Not implementation-focused
Tussyadiah [19]	2020	AI in tourism review	Confirms personalisation gap	Survey only
Calvaresi [20]	2021	Systematic review of tourism chatbots	Confirms multilingual gap	Survey only
Chase [21]	2023	LangChain framework	Composable LLM apps	Adds dependency surface
Touvron [22]	2023	LLaMA 2 open-weight models	Cost-effective fallback	Lower quality than top closed models

Table 1: Comparison of Existing Approaches and Travel Yantra

4.8 Comparison of Existing Works

4.9 Summary

From the works reviewed above, the following observations directly motivate the design choices of Travel Yantra:

- Generic large language models, used alone, are not sufficient for real multi-constraint travel planning, as demonstrated by the TravelPlanner benchmark.
- Multi-agent architectures perform measurably better than single-LLM systems on travel planning, but published systems do not provide multilingual or voice support for Indian users.
- IndicTrans2 and MURIL together offer a credible foundation for translation and code-switched understanding across the five most-spoken Indian languages.
- Retrieval-Augmented Generation is the standard technique to keep cultural and historical answers grounded and free from hallucination.
- Whisper provides a credible multilingual speech recognition backbone, although accent-specific fine-tuning may still be required.
- Neural Collaborative Filtering, Dijkstra and A* together cover the recommendation and routing needs of the system.

These observations collectively justify the multi-agent, multi-LLM, multilingual and multi-channel design adopted in the rest of this report.

5 RESEARCH GAP AND RESEARCH CHALLENGES

5.1 Research Gap

5.1.1 No Unified AI Travel Platform for India

None of the systems reviewed in Section 4 combines AI itinerary generation, real-time booking, on-trip cultural guidance, multilingual interaction and multi-channel access (web, WhatsApp and voice) in a single product. Booking platforms such as MakeMyTrip and Booking.com handle one slice each, generic chat assistants such as ChatGPT and Gemini handle conversation but cannot book or operate in regional languages, and Google Maps offers navigation without any cultural narrative. The integrated platform that Travel Yantra targets simply does not exist today [1, 2].

5.1.2 Indic Language and Code-Switched Queries are Under-Served

Even though IndicTrans2 [5] and MURIL [16] have moved Indic language understanding forward significantly, no deployed travel system has integrated them for real conversational use. Indic voice interfaces in commercial products show fifteen to thirty percent higher Word Error Rate than their English counterparts, and code-switched queries (Hinglish, Kanglish) remain a known weakness across the field [14].

5.1.3 No Cross-Session User Memory in Existing Travel Assistants

Existing chat-based travel assistants begin every conversation from zero. They do not learn that a user prefers vegetarian food, budget hotels, or cultural sites over beaches. Travel Yantra fills this gap with a Supabase-backed user profile that synchronises across web, WhatsApp and voice channels, which is a practical gap rather than a purely technical one but matters strongly for user satisfaction.

5.1.4 Multi-Agent Frameworks Are Not Tuned for Indian Tourism

Frameworks such as AutoGen [6] and CrewAI [2] provide generic multi-agent scaffolding, but none of them is tuned for the Indian tourism context, with its festival calendars, regional cuisines, religious site customs and seasonal travel patterns. This is an important practical gap that affects the quality of generated itineraries.

5.2 Research Challenges

5.2.1 Coordinating Multiple LLMs Without Latency or Context Loss

The system uses GPT-4, Claude 3.5 Sonnet and Gemini 1.5 Pro across different agents. Coordinating four agents that may belong to three different model families, while keeping end-to-end response time within three seconds and without losing context between hand-offs, is the central engineering challenge of the project.

5.2.2 Grounding Cultural Guidance Within a Tight Latency Budget

The Cultural Guidance Agent uses Retrieval-Augmented Generation [13] over a multilingual knowledge base. Retrieval, generation and translation must all complete within roughly three seconds, which is a non-trivial budget for a multi-stage pipeline.

5.2.3 Unified User Profile Across Web, WhatsApp and Voice

A user may begin a conversation on WhatsApp, switch to the web application later, and call in via voice during the trip itself. Designing a user profile and session schema that synchronises preferences, history and consent across these channels in real time, with Row-Level Security in Supabase, is a clear design challenge.

5.2.4 Multilingual Evaluation and Bias

Evaluating itinerary quality, translation accuracy and cultural appropriateness uniformly across English, Hindi, Tamil, Telugu, Kannada and Bengali requires careful per-language test design. This is necessary not only for engineering rigour but also to prevent English-centric bias in the final product.

5.2.5 Cost Discipline at Scale

The project must deliver all of the above within a twenty-five thousand rupee budget for Phase 1 and a sustainable per-user cost in production. This forces aggressive use of free tiers, open-source components, caching and the open-weight LLaMA 2 [22] as a fall-back, and is a constraint-driven challenge as much as a technical one.

6 PROPOSED METHODOLOGY

6.1 Overview

Travel Yantra is designed as a unified AI-powered travel ecosystem that spans the entire trip lifecycle: pre-trip planning, real-time booking, on-trip cultural guidance and post-trip preference learning. The user submits a request through one of three channels (web, WhatsApp or voice), which is dispatched by an n8n orchestrator to four specialised agents working in parallel. Each agent is paired with the LLM that is strongest at its sub-task and the agents share a common user profile and conversation history through a Supabase backend. The final response is composed, validated against the user's constraints, translated through IndicTrans2 if needed and delivered through the same channel the user came in on.

6.2 System Architecture

The architecture is organised in five layers, summarised below.

6.2.1 Input Layer

The input layer collects the user's request and standardises it into an internal request schema. The web client is built on React.js, the messaging client uses the WhatsApp Business API and the voice client uses Vapi.ai with Whisper as the underlying speech recognition model. All requests are authenticated using JWT tokens issued by Supabase and persisted to the user's session record.

6.2.2 Orchestration Layer

The orchestration layer is implemented in n8n. It receives the standardised request, decides which agents need to be invoked, dispatches them in parallel where possible, handles retries and fall-backs when a downstream API is unavailable, and merges the final response. The orchestration layer also exposes hooks for caching and rate limiting against the closed LLM APIs.

6.2.3 Agent Layer

The agent layer contains four specialised agents:

- **Itinerary Agent.** Built on GPT-4 with Chain-of-Thought [15] and ReAct [7] prompting. It performs constraint satisfaction over budget, dates, group composition and travel type, and uses Dijkstra [12] routing to construct day-by-day plans.

- **Accommodation Agent.** Built on Claude 3.5 Sonnet, with a Neural Collaborative Filtering [8] layer over hotel and homestay data. It calls live hotel APIs to retrieve real availability and pricing.
- **Transportation Agent.** Built on Gemini 1.5 Pro for its strength on real-time multimodal data, with A* search [18] for multi-stop scheduling. It calls flight and IRCTC APIs for live pricing.
- **Cultural Guidance Agent.** Built on Claude 3.5 Sonnet with Retrieval-Augmented Generation [13] over a multilingual cultural knowledge base. It uses IndicTrans2 [5] to translate the final answer into the user's preferred language.

6.2.4 Data and Backend Layer

The data layer uses Supabase, which provides a managed PostgreSQL instance with real-time subscriptions, JWT-based authentication and Row-Level Security. Row-Level Security ensures that each user can read and write only their own rows. User profiles, itinerary history, preference vectors and consent records all live here. Secrets such as model and API keys are stored in environment variables, never in source code.

6.2.5 Output Layer

The output layer composes the final itinerary from the agent outputs, embeds bookable links for transport and accommodation, attaches map embeds, attaches cultural notes, applies translation if required and ships the response back through the same channel the user came in on.

6.3 Agent to Model Mapping

The choice of LLM per agent is summarised below.

Agent	**LLM**	**Justification**
Itinerary Agent	GPT-4	Strongest constraint-satisfaction reasoning, well established Chain-of-Thought and ReAct behaviour.
Accommodation Agent	Claude 3.5 Sonnet	Long context for long hotel descriptions and reviews; structured output discipline.
Transportation Agent	Gemini 1.5 Pro	Strong on real-time multimodal data lookup against flight and rail APIs.
Cultural Guidance Agent	Claude 3.5 Sonnet	Strong narrative quality and long context, paired with RAG for grounding and IndicTrans2 for translation.

Table 2: Agent to Model Mapping and Justification

6.4 Data Sources and Preprocessing

Travel Yantra draws on several public and commercial data sources:

- Points of interest from OpenStreetMap and the Google Places API.
- Flight pricing from public flight aggregator APIs and rail data from IRCTC.
- Multilingual corpora from IndicCorp and Samanantar for fine-tuning the translation pipeline on travel vocabulary.
- User interaction logs (with explicit consent) for preference learning.

Preprocessing uses the IndicNLP Library for tokenisation and normalisation of Indic text, plus Named Entity Recognition for extracting locations, dates, budget figures and group composition from natural language queries.

6.5 Algorithms

The complete algorithmic stack is as follows:

- **LLM backbone:** GPT-4 for the Itinerary Agent, Claude 3.5 Sonnet for the Accommodation and Cultural Guidance agents, Gemini 1.5 Pro for the Transportation Agent, all with Chain-of-Thought [15] and ReAct [7] prompting.
- **Recommendation:** Neural Collaborative Filtering [8] on hotel and attraction profiles.
- **Routing:** Dijkstra [12] as baseline, A* [18] for multi-stop scheduling.
- **Translation and code-switching:** IndicTrans2 [5] for translation, MURIL [16] for code-switched understanding.
- **Retrieval and voice:** RAG [13] over a multilingual cultural knowledge base, Whisper [14] for speech recognition.

6.6 End-to-End Request Flow

The end-to-end flow when a user submits a request is summarised below.

1. The user submits a request through web, WhatsApp or voice. The request is authenticated via JWT and stored in Supabase.
2. The n8n orchestrator dispatches the request, in parallel, to the four agents.

3. The Itinerary Agent runs Chain-of-Thought planning with constraint satisfaction and Dijkstra or A* routing. The Accommodation and Transportation agents query their live APIs in parallel. The Cultural Guidance Agent queries its RAG store.
4. The agent outputs are merged, validated against the user's budget and time constraints, translated through IndicTrans2 if needed and persisted back to Supabase.
5. The final itinerary is delivered to the user through the original channel, with embedded booking links, a map embed and any required cultural notes.

6.7 Tools and Technologies

Layer	Tool	Role
Frontend	React.js	Web user interface for the planner.
Backend	FastAPI (Python 3.11)	REST endpoints, agent invocation, validation.
Orchestration	n8n	Multi-agent workflow orchestration and integrations.
LLMs	OpenAI GPT-4, Anthropic Claude 3.5 Sonnet, Google Gemini 1.5 Pro	Reasoning, narrative and real-time data agents.
NLP	IndicTrans2, MURIL, IndicNLP	Translation, code-switched understanding, tokenisation.
Speech	Whisper, Vapi.ai	ASR and voice channel.
Messaging	WhatsApp Business API	Conversational access for over five hundred million WhatsApp users in India.
Database	Supabase (PostgreSQL)	Profiles, history, RLS, JWT auth.
Maps	Google Maps Platform	Map embeds and routing data.
Open-weight fall-back	LLaMA 2 [22]	Cost-controlled fall-back for closed LLMs.

Table 3: Tools and Technologies used in Travel Yantra

6.8 Evaluation Plan

Evaluation is split across the three quality concerns of the system.

- **Itinerary quality:** Constraint Satisfaction Rate and Commonsense Violation Rate against a hand-curated test set of one hundred trips.
- **Translation quality:** BLEU [11] and METEOR [17] on a held-out travel-domain test set per language.
- **System performance:** P95 response latency, uptime percentage and a one-to-five Likert scale user satisfaction score collected from beta users.

6.9 Privacy, Standards and Sustainability

Standards followed include IEEE 830-1998 for the Software Requirements Specification, IEEE 12207 for life-cycle processes, IEEE 829 for test documentation and ISO/IEC 25010 for product quality. Data handling follows the Indian Digital Personal Data Protection Act 2023, with explicit consent at sign-up, data minimisation, user-initiated deletion and per-user Row-Level Security in Supabase. The stack is open-source-first (n8n, Supabase, React.js, IndicTrans2, MURIL, LangChain), which reduces both vendor lock-in and operational cost. Caching, batched inference and route optimisation jointly reduce both compute cost and on-trip emissions, and contribute to the project's sustainability story.

7 HIGHLIGHTS FROM PHASE 1 REVIEW 1 SUGGESTIONS, ALONG WITH JUSTIFICATION FOR INCLUSION

The Phase 1 Review 1 panel commended the team on a clear and well-structured presentation. Specific appreciation was noted for the depth of the literature survey, the precision of the problem identification and the fact that the objectives were measurable and time-bound. No specific corrections or improvement suggestions were raised by the panel. Even so, the team carried a number of self-initiated refinements into Phase 1 Review 2, which are recorded here for completeness and traceability.

7.1 Expanded Literature Survey

The literature survey was expanded from twelve papers in Review 1 to twenty-two IEEE-format papers in Review 2. The newly added papers cover Retrieval-Augmented Generation, Whisper, Chain-of-Thought prompting, MURIL, METEOR, the A* algorithm, LangChain, LLaMA 2 and AI-in-tourism surveys. The intent was to attach documented prior work to every component of the proposed architecture, rather than just to the headline LLM and translation pieces.

7.2 Finalised Multi-LLM Strategy

In Review 1 the team had noted that the choice between GPT and Claude was undecided. In Review 2 this is finalised as a multi-LLM strategy: GPT-4 for the Itinerary Agent, Claude 3.5 Sonnet for the Accommodation and Cultural Guidance agents, and Gemini 1.5 Pro for the Transportation Agent, all routed through n8n. The justification is that each agent's task plays to a different model's strength and the orchestration layer abstracts the routing.

7.3 Explicit Research Challenges Sub-Section

Review 1 grouped Research Gap and Research Challenges together. Review 2 separates them, so that the gap (what is missing in the literature and the market) is distinct from the challenges (what Travel Yantra has to solve technically). This makes the report easier to follow and easier to evaluate.

7.4 Standards, Sustainability and Project Management

Review 2 adds explicit coverage of engineering standards (IEEE 830, 12207, 829, ISO/IEC 25010), data-privacy practices (DPDP Act 2023, RLS, JWT, environment-variable secrets), ethics and responsible AI usage and a Gantt-style project management view. These were newly required by the Review 2 template and are now integrated into Section 6 and the project-management view at the end of this report.

7.5 Continuity Statement

The core problem statement, target users, system architecture and objectives remain unchanged from Review 1. The panel's positive feedback validated this baseline. Review 2 deepens the technical foundations and adds implementation evidence, without changing the project's direction.

8 EXPECTED OUTCOMES

The expected outcomes of the project are split across performance, innovation and mandatory project deliverables.

Performance Expectations. A fully functional multi-platform travel planning system, accessible through web, WhatsApp and voice, with at least ninety-five percent uptime and at least ninety percent user satisfaction. A sixty percent reduction in user planning time compared to traditional fragmented workflows. Coverage of six languages (English plus Hindi, Tamil, Telugu, Kannada and Bengali) with translation accuracy of eighty-five percent or higher and BLEU greater than twenty-five. Sub-three-second P95 response time for all query types and sub-one-second response time for simple conversational queries.

Innovation Contribution. A novel integration of n8n, Vapi voice AI, the WhatsApp Business API and a multi-LLM agent strategy, all tuned for the Indian multilingual travel market. To the best of the team's knowledge, this is the first open-architecture AI travel system supporting five Indian regional languages on a unified, multi-channel platform.

Mandatory Project Outcome. The primary outcome is a deployable, production-ready PRODUCT: a multi-channel AI travel planning application. The secondary outcome is a research publication targeted at IEEE or ACM proceedings on multi-agent AI for multilingual applications. Patent filing is under consideration but not committed.

Societal and Environmental Impact. The project aligns with five UN Sustainable Development Goals: SDG 8 by directing travellers to local homestays and small businesses, SDG 9 by demonstrating advanced multi-agent AI applied to a real industry problem, SDG 10 by removing the English-only barrier for over seven hundred million regional language speakers, SDG 11 by targeting a twenty percent reduction in transit time through route optimisation and SDG 17 through cross-sector partnerships with WhatsApp, Google Maps and government tourism portals.

Cost and Feasibility. The Phase 1 budget remains within twenty-five thousand rupees, allocated across LLM API access, voice and messaging APIs, cloud hosting, development tools and testing. An estimated forty thousand rupees in licensing is saved by using open-source components. Production cost is projected at approximately three thousand five hundred rupees per one thousand active monthly users, which supports a freemium commercial model.

9 CONCLUSION

This report has presented Travel Yantra, an AI-powered, multilingual, multi-channel travel itinerary planner and virtual tour guide for the Indian tourism context. The work begins from a clearly evidenced problem: travel planning in India is fragmented across four to five disconnected platforms and excludes over seven hundred million regional language users from the benefits of modern conversational artificial intelligence. The TravelPlanner benchmark of Xie et al. shows that even GPT-4, used alone, achieves less than one percent pass rate on complex multi-constraint travel plans, which confirms that a generic single-LLM approach is not sufficient for this task.

In response, twenty-two recent IEEE-format papers were surveyed across multi-agent LLM planning, multilingual NLP, RAG, voice ASR, recommendation, routing and tourism technology, and the resulting design choices were documented in Section 4. The proposed system combines four specialised agents on n8n, with each agent paired to the LLM strongest at its sub-task, supported by IndicTrans2 and MURIL for multilingual handling, RAG for grounded cultural answers, Whisper and Vapi for voice, the WhatsApp Business API for messaging and a Supabase backend with Row-Level Security for persistent user profiles.

Phase 1 Review 1 received positive panel feedback with no specific corrections, and Review 2 carries that baseline forward with an expanded literature survey, a finalised multi-LLM strategy, explicit research challenges, formal standards and sustainability coverage and a Gantt-style project management view. The project is technically grounded, economically feasible within the twenty-five thousand rupee budget and operationally viable from day one through WhatsApp's reach to over five hundred million users in India.

The remaining work in Phase 2 covers the Vapi voice channel, the Cultural Guidance RAG agent, full Maps and booking integration, beta testing across the five supported regional languages and production deployment. With the foundation completed in Phase 1, the path to a deployable product by the end of June 2026 is clear and on track.

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